

AI Academy, National AI Center  
AI-ENG-110 – Software Engineering

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# Async Research Assistant

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VerteX

Yaqub Xəlilli   Səbuhi Xamiyev   Aqil Əsgərov

[github.com/yagubbx/Research-Assistant](https://github.com/yagubbx/Research-Assistant)

# The research contract

## User outcome

A question becomes a concise answer with inspectable numeric references and visible missing-source notes.

### Input

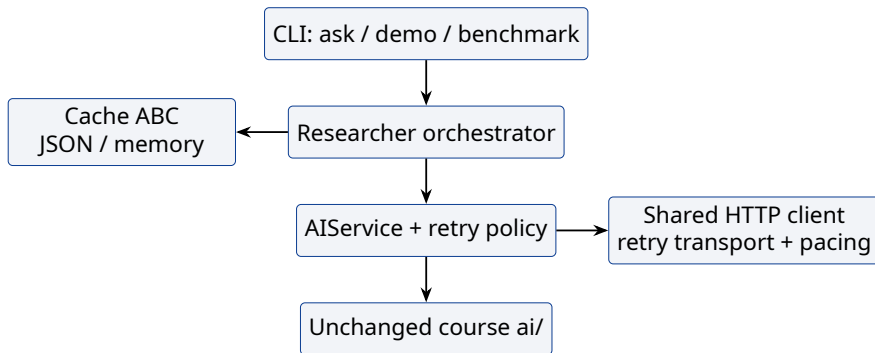
- ▶ Question: 1–2,000 characters
- ▶ Selected sources: wiki, arxiv, web
- ▶ Optional cache bypass

### Output

- ▶ Cited answer and source URLs
- ▶ Per-source status and timing
- ▶ Plain text or structured JSON

```
python -m researcher ask "your question"
```

# Architecture



Pydantic models cross boundaries. Only the service and its worker call the supplied AI package. No changes to `ai/`.

# Design decisions

## 1. **Async I/O for retrieval**

Independent sources overlap; a semaphore caps active operations.

## 2. **Filesystem cache behind an interface**

Persistence without a database server; memory backend supports tests.

## 3. **A process for synchronous synthesis**

A deadline can terminate the SDK call; startup adds latency.

## Retrieval benchmark

### Five questions, three repetitions, cache bypassed

|                | Sequential | Parallel |
|----------------|------------|----------|
| Median seconds | 2.412      | 0.802    |
| Speedup        | 1.00×      | 3.01×    |
| Failed / empty | 0          | 0        |

Synthetic HTTP delays: 150 ms per source; semaphore = 3.

**Scope:** retrieval only; no live-provider latency claim.

**Bottleneck:** the slowest source after overlap; synthesis may dominate the full live run.

## Integration problems and solutions

- ▶ **Uncited sentences:** validate and regenerate within the existing retry budget; never invent citations.
- ▶ **Empty or slow search:** simplify topic queries; bound web search in a killable worker.
- ▶ **Gemini errors:** inspect safe status codes; align model defaults with the working deployment.
- ▶ **Source failure:** keep usable evidence and show a missing-source note; stop if all sources fail.

Evidence: regression tests, Docker acceptance log and Render live sample.

## Scripted demonstration

```
python -m streamlit run researcher/web_ui.py
python -m researcher ask "photosynthesis" \
  --offline --sources wiki,arxiv --json
docker run --rm --network none vertex-research
```

Five original questions produce answers with numbered references.

Offline evidence is visibly labeled **SIMULATED**.

Render live sample: 23.21 s, 3 sources, 3 references. Offline mode needs no key.

## Testing evidence

**121** passing tests

**93.85%** application  
coverage

16 unchanged course smoke tests  
included.

Ruff passes; mypy reports no issues in 18 files. Tests use mocks and deny live sockets.

### High-value checks

- ▶ Source failure and timeout
- ▶ Semaphore bound and overlap
- ▶ 429 recovery; no 401 retry
- ▶ Cache expiry and corruption
- ▶ Invalid and missing citations

## Tested extensions in v1.1.0

- ▶ **Web failover:** primary failure or timeout, then a configured secondary; chaos tests verify recovery.
- ▶ **Token budget:** estimated sliding-window reservations; exhaustion test observes 60 s backoff.
- ▶ **Streamlit:** same core pipeline, sample/live modes, references and downloads; AppTest and browser checks.
- ▶ **CI:** lint, types, tests, coverage and container UI health workflow.

Docker acceptance passed. Live Gemini passed; PR 1 checks passed before merge. Bonus credit is assessed by the instructor.

## Team and AI disclosure

| Member         | Responsibility (team-reported)      |
|----------------|-------------------------------------|
| Yaqub Xəlilli  | CLI/core, integration and deploy    |
| Səbuhi Xamiyev | Providers, worker, retry and search |
| Aqil Əsgərov   | Cache, validation, offline and UI   |

Codex substantially drafted the software, tests and documentation.

Team reports about 90% joint sessions; implementation commits centralized under Yaqub. Filled contribution record awaits member signatures.

**Defense goal:** every member can explain the full request path.

# Questions

VerteX

`github.com/yagubbx/Research-Assistant`

Architecture, timeout behavior, citation validation and benchmark scope